Calculus with Parametric and Polar Curves

Parametric derivatives and arc length

1 For the curve $x = t^2$, $y = t^3$, find $\frac{dy}{dx}$ in terms of t.

2 Find the arc length of the curve $x = 3t^2$, $y = 2t^3$ for $0 \le t \le 2$.

Polar area

3 Find the area enclosed by one loop of the rose $r = 3\sin 2\theta$ (for $0 \le \theta \le \pi/2$).

4 Find the area enclosed by the cardioid $r = 2 + 2\cos\theta$.

Second derivatives and concavity for parametric curves

5 For $x = t^2$, $y = t^3 - 3t$, find $\frac{d^2y}{dx^2}$ in terms of t, using $\frac{d^2y}{dx^2} = \left[\frac{d}{dt}\left(\frac{dy}{dx}\right)\right] \div \frac{dx}{dt}$.

Area between polar curves

6 Find the area inside the circle $r = 3$ and outside the circle $r = 2$ (an annular region).

Horizontal and vertical tangents on parametric curves

7 For $x = t^2 - 4$, $y = t^3 - 3t$, find the value(s) of t where the tangent line is horizontal ($\frac{dy}{dt} = 0$, $\frac{dx}{dt} \neq 0$).

8 For the same curve, find the value(s) of t where the tangent line is vertical ($\frac{dx}{dt} = 0$, $\frac{dy}{dt} \neq 0$).

Converting between polar and Cartesian for calculus

9 Find the slope of the tangent line to $r = 1 + \sin\theta$ at $\theta = 0$, using $x = r\cos\theta$, $y = r\sin\theta$ as parametric equations in θ.

Speed and length for parametric motion

10 A particle moves according to $x(t) = t^2$, $y(t) = 4t$. Find its speed at $t = 3$.

11 Find the arc length of $x = \cos^3 t$, $y = \sin^3 t$ for $0 \le t \le \frac{\pi}{2}$ (an astroid arc), using $\frac{dx}{dt} = -3\cos^2 t\sin t$ and $\frac{dy}{dt} = 3\sin^2 t\cos t$.

Slope and concavity for polar-derived curves

12 Find $\frac{dy}{dx}$ for the polar curve $r = 2\theta$ at $\theta = \frac{\pi}{2}$, using $x = r\cos\theta$, $y = r\sin\theta$.

13 For $x = t^3 - 3t$, $y = t^2$, find the points where the tangent is horizontal.

Arc length of a polar curve

14 Set up (do not evaluate) the arc length integral for the circle $r = 4$ over $0 \le \theta \le 2\pi$, using $L = \int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$, and verify the setup is consistent with the known circumference formula.

Applied parametric motion

15 A projectile's position is given by $x(t) = 20t$, $y(t) = 15t - 4.9t^2$ (meters, seconds). Find the projectile's velocity components $\frac{dx}{dt}$ and $\frac{dy}{dt}$, and find the speed at $t = 1$.

16 For the same projectile, find the time at which the vertical velocity is 0 (the peak of the trajectory).

Polar curves and symmetry

17 The cardioid $r = 1 + \cos\theta$ is symmetric about the polar axis. Use this symmetry to explain why the total area enclosed can be computed as $2\times$ the area for $0 \le \theta \le \pi$, and evaluate the total area.

18 For the parametric curve $x = e^t\cos t$, $y = e^t\sin t$, find $\frac{dy}{dx}$ at $t = 0$, using $\frac{dx}{dt} = e^t(\cos t - \sin t)$ and $\frac{dy}{dt} = e^t(\sin t + \cos t)$.

19 A Ferris wheel of radius 10 m, centered 12 m above the ground, rotates so that a rider's position is $x(t) = 10\cos t$, $y(t) = 12 + 10\sin t$, where t is measured in radians and represents elapsed angle since the rider was at the rightmost point. Find the rider's height above the ground when $t = \frac{\pi}{3}$, and find the vertical velocity $\frac{dy}{dt}$ at that instant (assuming t increases at a constant rate of 1 rad per unit time).

Total distance traveled along a parametric path

20 A robot on a factory floor follows the path $x(t) = t^2$, $y(t) = 2t$ for $0 \leq t \leq 3$ (meters, seconds). Set up the arc length integral for the total distance traveled, and evaluate it exactly using the substitution $u = t^2 + 1$ where appropriate.